

Data-driven collaborative optimization of three-dimensional highway alignments for life-cycle cost and mixed fuel–electric traffic energy

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Abstract

Horizontal and vertical highway alignment decisions jointly determine terrain disturbance, structures, construction expenditure and long-term vehicle operation. This study develops a spatially explicit bi-objective framework that combines 14,018 GNSS centerline points, a quasi-natural digital elevation model reconstructed after masking the existing roadway, and OpenStreetMap transport and water features. Candidate plans are represented by 50 sine modes, whereas one start-elevation variable and 224 segment grades describe the profile. Life-cycle cost covers land, basic works, structures, earthwork and maintenance; the second objective monetizes 30-year fuel and electricity use for a mixed fleet. Intersections with qualifying OSM features trigger bridge intervals, a connected mountain zone triggers ecological tunnelling, and Improved Jellyfish Search (IJS) approximates the cost–energy front. Feasibility, a planning budget, non-dominance and entropy weighting are applied sequentially to select a compromise.

For the 22.46-km Guangzhou North Ring case, the feasible joint solution reduces cost from 2.6406 to 2.4224 billion RMB (8.26%) and monetized energy from 1.3946 to 1.3477 billion RMB (3.37%). It shortens the line by 98.3 m and satisfies the 400-m radius requirement with $R_{min} = 401.0$ m and zero stored penalty. Structure cost decreases by 261.4 million RMB, offsetting a 45.5-million-RMB earthwork increase. A sequential solution has lower raw objectives but is infeasible ($R_{min} = 397.1$ m). Same-generation ablation and algorithm tests show the lowest observed mean objective for IJS, although unequal function-evaluation counts preclude an equal-budget superiority claim. The unanchored vertical endpoints and single-direction energy calculation limit the result to corridor-planning evidence.

Keywords: highway alignment optimization, digital elevation model, OpenStreetMap, improved Jellyfish Search, life-cycle cost, traffic energy, horizontal-vertical coordination

1. Introduction

Highway alignment is an early decision with long-lived consequences. A lateral shift changes the terrain crossed, land occupation, bridge and tunnel demand, earthwork balance and grades experienced by vehicles. These effects are coupled: a shorter plan may worsen the profile, avoiding a mountain may create additional crossings, and a locally cheap profile may become geometrically infeasible. Alignment selection is therefore a constrained spatial and multi-objective problem rather than separate curve fitting in plan and profile.

Classical formulations established the value of simultaneous horizontal and vertical optimization [1, 2]. Later work introduced GIS corridors, mixed-integer models, environmental objectives and preference-aware search [3, 4, 5, 6, 7]. Recent studies combine alignment generation with BIM, reinforcement learning, sampling-based planning and hybrid metaheuristics [8, 9, 10, 11, 12]. Nevertheless, heterogeneous spatial data and traceable engineering accounting remain difficult to integrate consistently [13].

Three gaps motivate this study. First, a single centerline terrain profile cannot reflect the terrain encountered after horizontal displacement. Second, fixed bridge and tunnel allowances may allow the optimizer to exploit accounting assumptions instead of locating a constructible corridor. Third, high-dimensional plan-profile search needs both global exploration and local refinement, while feasibility must be checked at fine spatial resolution.

This paper contributes:

- (1) a digital corridor linking each candidate plan to a quasi-natural DEM and OSM crossing features;
- (2) an auditable life-cycle formulation with geometry-triggered bridges, ecological tunnelling and earthwork-to-structure substitution;
- (3) a 275-variable collaborative plan-profile representation and a cost-energy decision sequence that places feasibility before entropy weighting; and

(4) a boundary-aware evaluation of IJS, including the engineering solution, a joint-versus-sequential control, same-generation mechanism ablation, multi-algorithm tests and re-optimization sensitivity analysis.

Figure 1 summarizes the workflow. All numerical claims in this manuscript are fixed to repository commit `c1327df`; no later endpoint or density formulation is mixed into the evidence. The full hash is recorded in the accompanying source manifest.

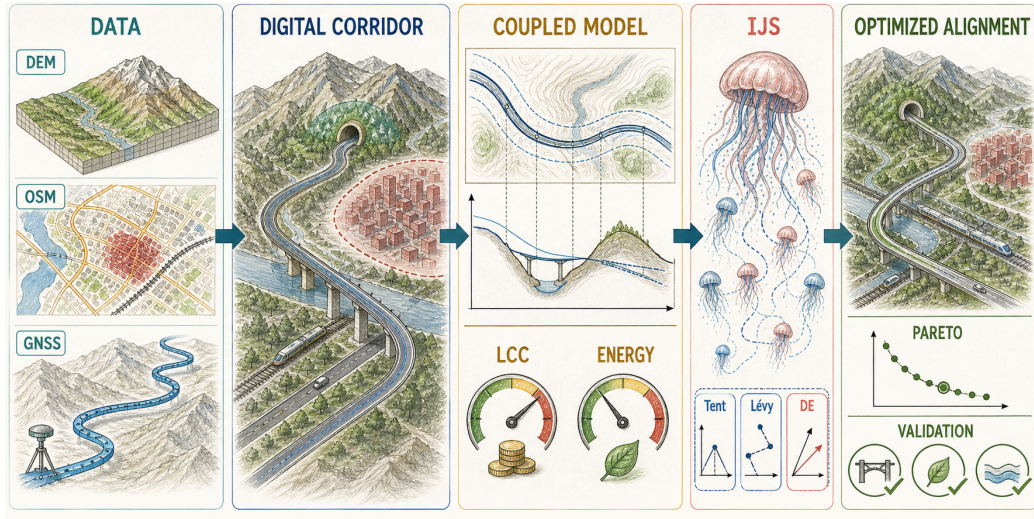


Figure 1: Data-driven workflow from spatial data and coupled objectives to IJS search and compromise selection. Buildings in the illustration provide corridor context; the pinned model uses DEM and OSM transport/water crossings but contains no building-density constraint.

2. Related research

2.1. Three-dimensional alignment optimization

Simultaneous plan–profile models avoid inconsistencies caused by sequential design [1, 2]. GIS-based methods broadened the feasible corridor and incorporated environmental layers [3, 14]; corridor-constrained and mathematical-programming approaches improved tractability [4, 7]. Bi-objective highway and railway models further introduced cost, hazard and ecological performance [5, 15, 16]. The present study emphasizes a traceable mapping from candidate coordinates to terrain, crossings, quantities and objectives.

2.2. Life-cycle cost and traffic energy

The cost categories follow the mathematical base in Lin [17]: land, bridge/tunnel works, basic works, maintenance and earthwork. Road geometry affects vehicle energy through distance, grade and curvature [18]; electric-vehicle consumption also depends on drivetrain efficiency and regenerative operation [19]. Fuel and electricity are therefore retained separately and then monetized over a common 30-year horizon. Monetary aggregation provides a decision scale but is not a complete environmental assessment.

2.3. Search and compromise selection

NSGA-II is a standard multi-objective benchmark [20]. Jellyfish Search (JS) alternates ocean-current and active/passive movement [21]. The implemented IJS adds Tent initialization, Lévy exploration and differential-evolution (DE) refinement [22]. Entropy weighting, derived from Shannon information [23], is used twice: first to establish a common scalar search direction and later to select among feasible non-dominated candidates. The two uses are kept distinct.

3. Data-driven mathematical formulation

3.1. Study corridor and spatial data

The case covers the Xunfengzhou–Huangcun section of Guangzhou North Ring Expressway, a two-way six-lane urban expressway. The measured M-A centerline contains 14,018 GNSS points and is 22.4618 km long under the common evaluator. All candidates share the measured plan endpoints.

Terrain is obtained from 60 AWS Terrain Tiles at zoom level 14, approximately 8.8 m per pixel [24]. To reduce the influence of the built road surface, a ± 60 -m mask around the existing road is removed and interpolated from adjacent terrain, producing a quasi-natural DEM. After a constant vertical-datum adjustment, the reconstructed surface has a correlation of 0.915 with measured centerline elevation. OSM supplies road, railway and water features [25, 26]. After removal of the North Ring Expressway and parallel facilities, the stored obstacle layer contains 4,807 road, 857 railway and 338 river/canal features.

The DEM is queried at the actual coordinates of each candidate line. Intersections with motorway, trunk or primary roads, railways and water-courses trigger bridge intervals. A connected high-terrain region represents

81 the Baiyun ecological tunnel zone. OSM buildings are available only as visual
 82 context in this pinned version and do not enter the objective or constraints.

83 3.2. Collaborative decision variables

84 Let the measured plan be $\mathbf{r}_0(s) = [x_0(s), y_0(s)]^\top$ with unit normal $\mathbf{n}(s)$,
 85 $s \in [0, L_0]$. A candidate normal offset is

$$\delta(s) = \sum_{k=1}^{50} a_k \sin\left(\frac{k\pi s}{L_0}\right), \quad |a_k| \leq \frac{W}{k^2}, \quad (1)$$

86 and $\mathbf{r}(s) = \mathbf{r}_0(s) + \delta(s)\mathbf{n}(s)$. The sine basis fixes both plan endpoints and
 87 attenuates high-frequency curvature. In the main run $W = 500$ m is the
 88 amplitude of the first mode, not a pointwise corridor half-width; the conser-
 89 vative envelope is $|\delta| \leq W \sum_{k=1}^{50} k^{-2} \approx 1.625W$. Cubic-spline reconstruction
 90 is evaluated at approximately 10-m spacing. Curvature and radius are

$$\kappa(s) = \frac{|x'(s)y''(s) - y'(s)x''(s)|}{[x'(s)^2 + y'(s)^2]^{3/2}}, \quad R(s) = \kappa(s)^{-1}. \quad (2)$$

The profile block has $M = 225$ normalized variables. With quasi-natural
 ground z_0^g , start-amplitude $A_0 = 20$ m, maximum grade $g_{max} = 0.04$, control
 spacing Δs_i and variables $u_i \in [0, 1]$,

$$z_0 = z_0^g + (2u_0 - 1)A_0, \quad (3)$$

$$g_i = (2u_i - 1)g_{max}, \quad i = 1, \dots, 224, \quad (4)$$

$$z_j = z_0 + \sum_{i=1}^j g_i \Delta s_i. \quad (5)$$

91 Thus the full search vector has 50 plan coefficients, one start-elevation vari-
 92 able and 224 grade variables, totaling 275 active variables. The grade bound
 93 is satisfied by construction. Crucially, Eq. (5) does not anchor either profile
 94 endpoint to the measured road. The implication is reported explicitly in
 95 Section 6.3.

96 3.3. Life-cycle cost

97 The first objective is

$$\min C = C_R + C_B + C_S + C_Q + C_{TU}, \quad (6)$$

where C_R , C_B , C_S , C_Q and C_{TU} denote land, bridge/tunnel, basic works, discounted maintenance and earthwork/haulage costs. With road width B , route length L , bridge and tunnel lengths L_B, L_T , and invariant interchange ramp cost C_{ramp} ,

$$C_R = c_{land} \frac{LB}{666.67} + 5000 \frac{L}{1000}, \quad (7)$$

$$C_B = c_B L_B + C_{ramp} + c_T L_T, \quad (8)$$

$$C_S = (c_{sg} + c_{pv})L, \quad (9)$$

$$C_Q = \sum_{y=1}^T \frac{1}{(1+r)^y} \sum_i \Delta s_i [\gamma + 365 Q_y \tau + c_{soil} \Phi_i]. \quad (10)$$

98 The analysis horizon is $T = 30$ years and $r = 5\%$. In the maintenance
 99 term, $\gamma = 10$ RMB/m is the traffic-independent annual rate, $\tau = 10^{-6}$
 100 RMB/(vehicle-trip m) is the traffic coefficient, and $c_{soil} = 2$ RMB/m weights
 101 slope exposure. Specifically, $\Phi_i = \alpha W_{s,i} m_s^2 + \beta W_{h,i} m_h^2 + (1 - \alpha - \beta)(W_{s,i} m_s^2 +$
 102 $W_{h,i} m_h^2)$, with $\alpha = \beta = 0.3$. Here $W_{s,i}$ and $W_{h,i}$ are the cut- and fill-slope
 103 widths, and m_s and m_h are their side-slope ratios; both widths are zero
 104 on structure-exempt segments. Unit rates follow the source thesis and the
 105 Guangzhou governmental estimation guideline [27].

106 At station j , $h_j = z_j - z_j^g$ and the approximate earthwork area is

$$A_j = B|h_j| + m h_j^2. \quad (11)$$

Cut and fill volumes use the average-end-area rule. For non-exempt stations, the code selects the lower of earthwork and the corresponding per-metre structure cost and forces structure substitution when $|h_j| > 30$ m. The selected station rates are then integrated trapezoidally:

$$V^\pm = \sum_j \frac{A_j^\pm + A_{j+1}^\pm}{2} \Delta s_j, \quad (12)$$

$$c_j = \begin{cases} c_{str}, & |h_j| > 30 \text{ m or } c_{ew} A_j > c_{str}, \\ c_{ew} A_j, & \text{otherwise,} \end{cases} \quad (13)$$

$$C_{TU} = E_L + \sum_j \frac{c_j + c_{j+1}}{2} \Delta s_j. \quad (14)$$

Table 1: Principal parameters of the pinned case model.

Group	Parameter	Value
Geometry	speed / width / R_{min} / g_{max}	100 km/h / 25.5 m / 400 m / 4%
Representation	plan modes / profile variables / W	50 / 225 / 500 m
Life cycle	horizon / discount rate	30 yr / 5%
Structures	bridge / tunnel / E_{ext}	156 / 270 million RMB/km / 75 m
Earthwork	cut / fill / side slope	30 / 25 RMB/m ³ / 1:1.5
Traffic	AADT / EV share	30,000 veh/day / 0.30
Energy	fuel / electricity price	8 RMB/L / 0.8 RMB/kWh
Search	population / iterations / weight points	200 / 1,000 / 21

107 Ecological-tunnel and crossing-bridge intervals are earthwork-exempt to pre-
 108 vent double counting. For an obstacle with nominal width b_q , crossing angle
 109 θ_q and approach extension $E_{ext} = 75$ m, its bridge interval is

$$\ell_q = \frac{b_q}{\max(\sin \theta_q, \sin 15^\circ)} + 2E_{ext}, \quad (15)$$

110 and gaps below 100 m are merged.

111 3.4. Mixed fuel-electric traffic energy

112 The second objective is the discounted monetary cost of fuel and electric-
 113 ity:

$$\min E = \sum_{y=1}^{30} \frac{365Q_y}{(1+r)^y} [(1 - p_{EV})c_f e_f(\mathbf{x}) + p_{EV}c_e e_e(\mathbf{x})]. \quad (16)$$

Here e_f is single-vehicle single-trip fuel use in L and e_e is electricity use in kWh. For a representative vehicle, aerodynamic, rolling, grade and residual lateral forces are

$$F_{air} = \frac{1}{2}\rho C_d A_f v^2, \quad F_r = C_r m_v g \cos \vartheta, \quad (17)$$

$$F_g = m_v g \sin \vartheta, \quad F_a = \max \left(0, \frac{m_v v^2 / R - m_v g S_p}{N_v C_s} \right). \quad (18)$$

The grade angle is $\vartheta = \arctan g_i$. The implemented superelevation is $S_p = \min(S_{p,max}, v^2/(gR))$, with $S_{p,max} = 0.08$. The 10^{-3} multiplier printed at the end of the source-thesis lateral-force equation is omitted in the pinned implementation because it is dimensionally and numerically inconsistent; the equation above reports the committed implementation. Here $\rho = 1.25$ kg/m³,

$C_d = 0.28$, $A_f = 2.0/2.2 \text{ m}^2$, $C_r = 0.015/0.012$, and $m_v = 1500/1800 \text{ kg}$ for fuel/electric vehicles; $N_v = 4$ and $C_s = 43$. In Eq. (20), $\phi = 0.03 \text{ ml}/(\text{s hp})$, $HP_{in} = 27 \text{ hp}$ and $\eta_f = 0.85$. Fuel use converts positive external load to horsepower:

$$P_{ex,i} = \max\left(0, \frac{(F_{air} + F_r + F_g + F_a)v}{736\eta_f}\right), \quad (19)$$

$$e_f(\mathbf{x}) = \sum_i \phi(HP_{in} + P_{ex,i}) \frac{\Delta s_i}{v} / 1000. \quad (20)$$

114 For electric vehicles, with signed work $W_i = (F_{air} + F_r + F_g + F_a)\Delta s_i$,

$$e_e(\mathbf{x}) = \left[\frac{1}{3.6 \times 10^6} \sum_i \begin{cases} W_i/\eta_e, & W_i \geq 0, \\ W_i\eta_e, & W_i < 0, \end{cases} \right]_+ + 0.15L_{km}, \quad (21)$$

115 where $[x]_+ = \max(x, 0)$ and $\eta_e = 0.90 \times 0.95$. The code evaluates one repre-
116 sentative travel direction; fuel and electricity components remain separately
117 reportable before aggregation.

118 3.5. Constraints and decision rule

119 The hard conditions in the pinned implementation are $R(s) \geq 400 \text{ m}$,
120 $|g_i| \leq 4\%$, adjacent grade-change limit $|g_{i+1} - g_i| \leq 3 \times 10^{-4} \Delta s_{ctrl}$ and
121 crossing skew angle $\theta \geq 15^\circ$. Plan endpoints and grade bounds are satisfied
122 by construction; vertical endpoints are free. Values follow JTG D20–2017
123 [28] where directly applicable.

124 For a relative violation vector $\mathbf{v}$, define $H(\mathbf{v}) = \text{mean}(\mathbf{v}) + \max(\mathbf{v})$. With
125 $v_R = [400 - R]_+/400$, normalized grade-change excess $v_{\Delta g}$ and skew excess
126 v_{skew} , the implemented penalty is

$$P = 10\rho_t[H(\mathbf{v}_R) + H(\mathbf{v}_{\Delta g}) + H(\mathbf{v}_{skew})], \quad (22)$$

127 where $\rho_t = 0.3$ in the first search half and 3.0 in the second. The mean-plus-
128 maximum form prevents a short severe violation from being diluted by route
129 length.

130 A common initial population supplies C_{ref} , E_{ref} and entropy weights
131 $(w_C, w_E) = (0.50838, 0.49162)$. Each search minimizes

$$F(\mathbf{x}) = w_C \frac{C(\mathbf{x})}{C_{ref}} + w_E \frac{E(\mathbf{x})}{E_{ref}} + P(\mathbf{x}), \quad (23)$$

132 with $C_{ref} = 4.04836$ billion RMB and $E_{ref} = 1.40373$ billion RMB. Twenty-
 133 one values of $w_C \in [0, 1]$ approximate the front. Candidates then pass four
 134 gates: $P \leq 10^{-6}$, $C \leq 1.1C_{M-A}$, non-dominance and entropy-weighted com-
 135 promise selection. Front entropy gives (0.41421, 0.58579); the selected candi-
 136 date originates from the $w_C = 0.50838$ search.

137 4. Improved Jellyfish Search

138 4.1. Mechanisms

139 The original JS alternates ocean-current motion toward the best solution
 140 with active/passive population movement [21]. The implemented IJS adds:

- 141 • independent Tent-map initialization ($\mu = 1.99$ in the actual algorithm);
- 142 • Lévy perturbations scaled by $0.01(\mathbf{u} - \mathbf{l})$ for non-local exploration; and
- 143 • DE mutation and crossover with $CR = 0.5$, followed by elitist acceptance
 144 [22].

145 At generation t , the JS time control is $A_r = (1 - t/T)(2u - 1)$. The
 146 implementation applies a JS trial, then a Lévy trial and finally a DE trial,
 147 accepting improvements after each stage. Consequently a full IJS generation
 148 uses approximately three population-equivalent objective-evaluation phases,
 149 whereas plain JS uses one. The soft-to-hard penalty schedule is applied across
 150 two consecutive half-runs. Figure 2 summarizes the implemented process.

151 4.2. Experimental design and statistical scope

152 The engineering experiment uses $W = 500$ m, 275 variables, population
 153 200, 1,000 generations and 21 weight points. M-A is the existing line, M-
 154 B minimizes cost alone, and M-C is the final feasible entropy-selected joint
 155 solution. A two-stage control first minimizes plan-related life-cycle cost and
 156 freezes the resulting plan; the second stage then optimizes the profile using
 157 the joint experiment’s common C/E references and entropy weights. Each
 158 stage runs for 500 generations.

159 The mechanism ablation uses a measured-plan, 225-variable profile
 160 testbed with 30 independent runs, population 200 and 500 generations.
 161 The multi-algorithm test compares IJS, JS, NSGA-II, real-coded GA, PSO
 162 and GWO at six profile resolutions with ten runs per algorithm–resolution
 163 pair. Ablation p values use the independent-sample Mann–Whitney U test,

Table 2: Existing and entropy-selected joint schemes at commit `c1327df`.

Metric	M-A existing	M-C joint	Change
Life-cycle cost (10^8 RMB)	26.4061	24.2239	−8.26%
Traffic energy (10^8 RMB)	13.9460	13.4766	−3.37%
Fuel component	11.1072	10.7512	−3.21%
Electricity component	2.8388	2.7254	−3.99%
Length (km)	22.4618	22.3635	−0.44%
Minimum radius (m)	422.43	401.03	feasible
Slope-hazard index	3.309	3.420	+3.36%
Penalty	0	0	feasible

Cost and energy are discounted 30-year monetary quantities. Changes are calculated from unrounded JSON values.

whereas benchmark p values use the independent Wilcoxon rank-sum test (SciPy `ranksums`); neither analysis is paired. Because activating Lévy and DE adds objective calls, equal generations do not imply equal numbers of function evaluations (NFE). These tests support observed same-generation differences, not equal-budget causal claims.

Sensitivity analysis contains 237 already-completed re-optimizations covering demand, fleet, energy prices, design parameters and spectral width. These data use the same free-endpoint model family as the pinned engineering result. No supplementary experiment is introduced in this manuscript.

5. Results

5.1. Engineering result

Table 2 and Figure 3 show the feasible joint solution. Cost decreases by 8.26% and monetized energy by 3.37%; route length decreases by 98.3 m (0.44%). Fuel and electricity components fall by 3.21% and 3.99%, respectively. The diagnostic slope-hazard index rises by 3.36%, so the result is not a simultaneous improvement in every reported quantity. This index is a screening diagnostic rather than an optimization objective or probability of failure.

The cost mechanism is dominated by structure substitution. Bridge/tunnel cost decreases by 261.4 million RMB, whereas earthwork cost increases by 45.5 million RMB; maintenance rises by 0.88 million RMB, yielding a net

Table 3: Joint and two-stage results under common objective references.

Metric	M-A	Two-stage	Joint M-C
Cost (10^8 RMB)	26.4061	23.1582	24.2239
Energy (10^8 RMB)	13.9460	13.4569	13.4766
Length (km)	22.4618	22.3659	22.3635
R_{min} (m)	422.43	397.06	401.03
Penalty	0	0.07359	0
Decision status	feasible	infeasible	feasible

cost reduction of 218.2 million RMB. Land and basic works also decrease slightly. Hence the saving is not attributable only to route shortening; it reflects a larger reduction in structures offset by greater earthwork.

The cost–energy sweep contains 21 weighted searches. Figure 4 shows the feasible points and the entropy-selected M-C. The feasibility and 10% budget gates remove inadmissible or structure-heavy endpoints before the final entropy score is evaluated.

5.2. Joint versus sequential optimization

The sequential control achieves lower raw values than the joint solution, but it violates the minimum-radius requirement. Table 3 therefore distinguishes numerical objective values from admissible engineering performance. Freezing the plan after stage one leaves the profile stage unable to repair the 397.06-m plan radius; its penalty remains 0.07359. The feasible M-C is the preferred result despite its higher raw cost and energy.

5.3. IJS mechanism ablation

At 500 generations, complete IJS has the lowest observed mean scalar objective, 0.6808 versus 0.9133 for JS, an observed gap of 25.45% (Table 4). Among the single-operator variants, JS+DE has the largest observed same-generation gap from JS; Tent alone has a small, statistically inconclusive gap in the unadjusted rank-sum test. IJS reaches the final JS level at generation 187 in the stored convergence record. However, the full method also uses about three times as many objective evaluations per generation. The table therefore demonstrates same-generation performance only.

Table 4: Thirty-run profile-testbed ablation. p values are unadjusted independent-sample rank-sum tests; NFE differs by variant.

Variant	Mean F	SD	F_{100}	Gap from JS	p
JS	0.9133	0.0637	2.547	—	—
JS+Tent	0.8989	0.0605	2.227	1.58%	0.464
JS+Lévy	0.8787	0.0558	2.095	3.79%	0.0555
JS+DE	0.6926	0.0315	1.679	24.17%	3.02×10^{-11}
IJS	0.6808	0.0340	1.292	25.45%	3.02×10^{-11}

Table 5: Objective mean (SD) over ten runs. Equal generations do not imply equal NFE.

Algorithm	PJ1 (95)	PJ2	PJ3	PJ4	PJ5	PJ6 (500)
IJS	.6617 (.0051)	.5986 (.0045)	.7087 (.0045)	.7090 (.0033)	.7778 (.0078)	.8333 (.0019)
JS	.6771 (.0052)	.6104 (.0049)	.7237 (.0063)	.7215 (.0066)	.7954 (.0020)	.8563 (.0060)
NSGA-II	.6792 (.0045)	.6240 (.0057)	.7282 (.0053)	.7287 (.0080)	.8104 (.0084)	12.8859 (.7330)
GA	.7103 (.0088)	.6478 (.0171)	.7515 (.0090)	.7489 (.0071)	5.3906 (.5481)	25.6024 (1.0630)
PSO	.7023 (.0116)	.6370 (.0113)	.7436 (.0070)	.7632 (.0072)	9.1988 (.9109)	32.7437 (.6336)
GWO	.6843 (.0065)	.6194 (.0070)	.7233 (.0046)	.7276 (.0070)	1.1104 (.0158)	3.2157 (1.5706)

5.4. Multi-algorithm comparison

IJS has the lowest recorded mean F at all six resolutions (Table 5). At PJ6, IJS and JS return feasible solutions in all ten runs, whereas the four other methods return penalty-dominated solutions. IJS takes about 1.9 times the wall time of JS at PJ1–PJ5 and about 2.8 times at PJ6, while also performing more objective calls. The defensible conclusion is therefore that the implemented multi-phase search attains the lowest observed same-generation objective and stronger high-resolution feasibility in this record, not that it is superior under equal computational budget.

5.5. Sensitivity within the pinned model family

All 237 stored sensitivity re-optimizations are marked feasible under their implemented checks. Traffic growth dominates the monetary energy response; cost is less responsive to demand, fleet and energy-price factors because most infrastructure expenditure occurs at construction. Higher EV penetration moderates the high-demand energy burden, whereas combined fuel and electricity price growth raises E .

The spectral-width response is strong and then saturating: optimized C decreases from 2.568 billion RMB at $W = 200$ m to 2.239 billion RMB at $W = 2,500$ m. The result supports the interpretation of W as search

227 flexibility, not a hard corridor width. Changing E_{ext} from 50 to 75 and 100
228 m shifts C to 2.281, 2.355 and 2.474 billion RMB, while E stays near 1.355
229 billion RMB. These values show that bridge assumptions materially affect
230 economic results.

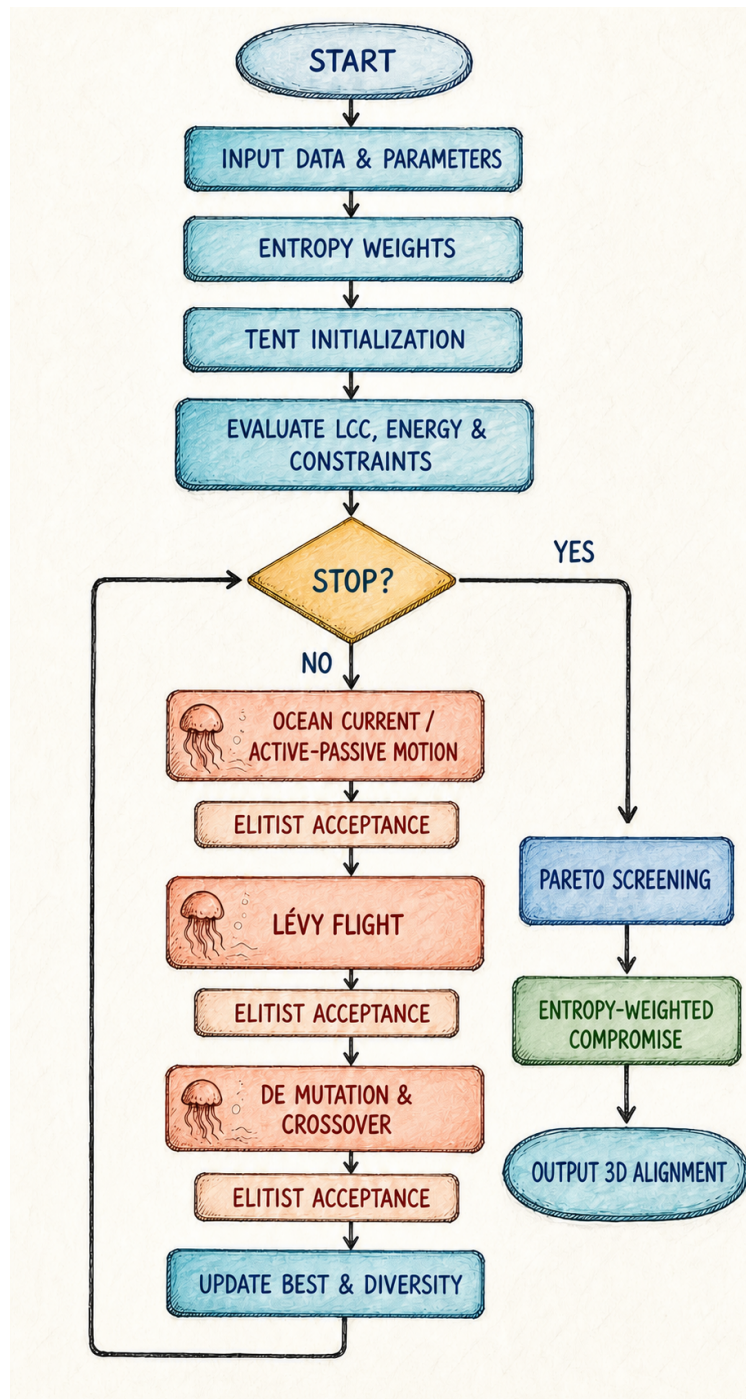


Figure 2: IJS search and post-search decision sequence.

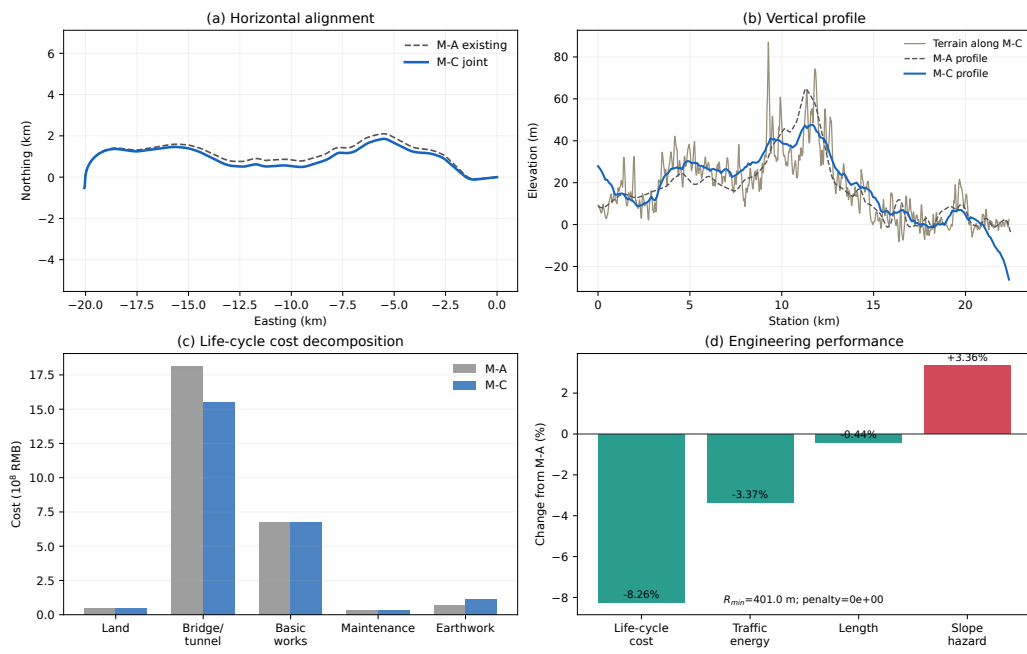


Figure 3: Pinned M-A/M-C comparison: plan, profile, cost decomposition and headline changes. The unanchored terminal profile is shown rather than concealed.

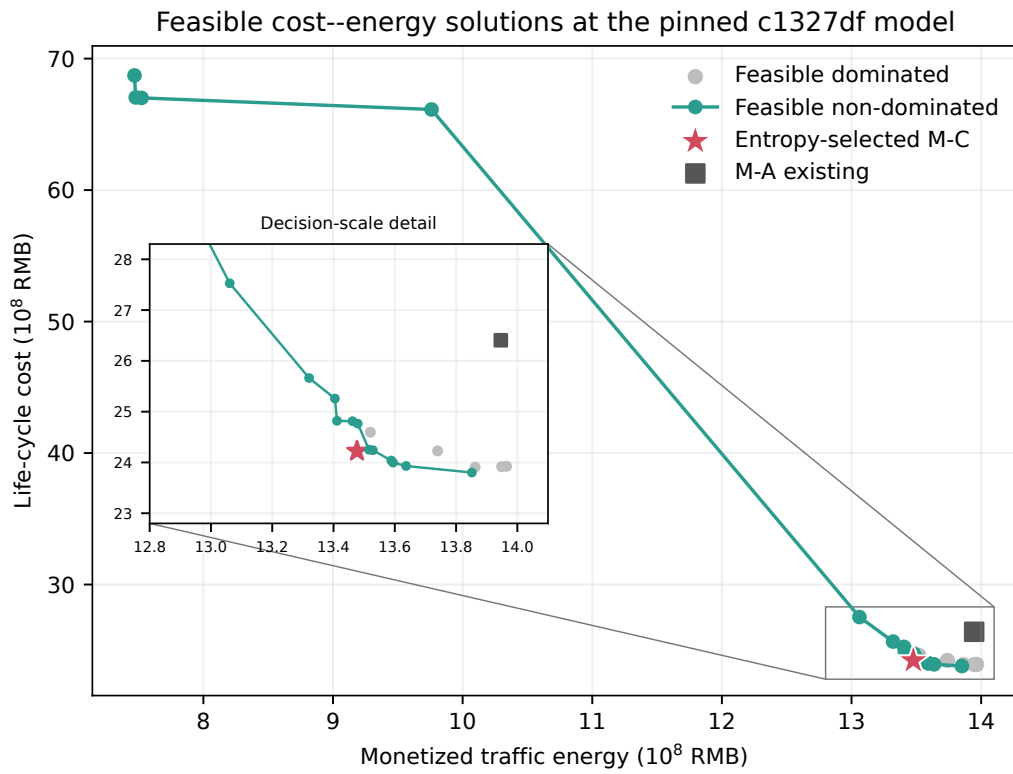
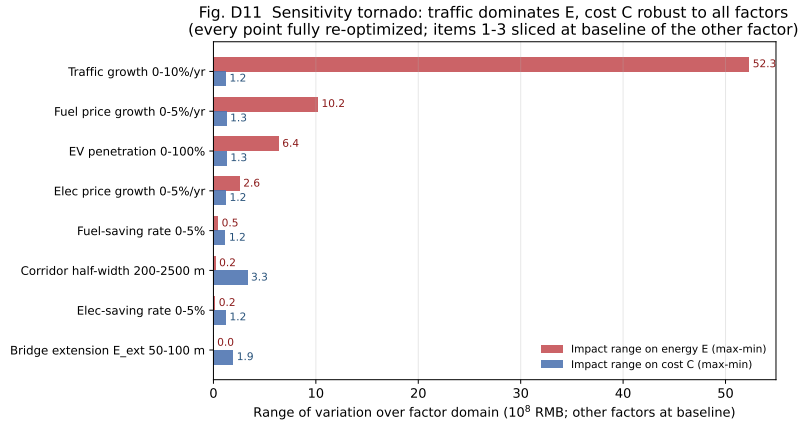
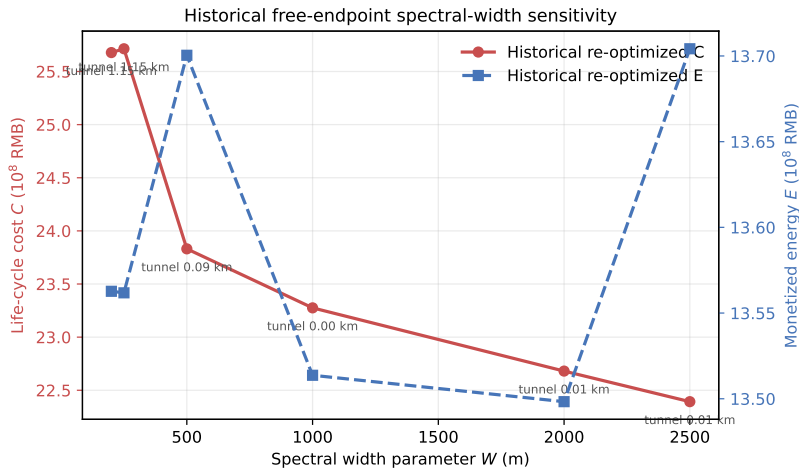


Figure 4: Feasible cost-energy solutions and the entropy-selected compromise in the pinned model.



(a) Relative importance of operating scenarios.



(b) Re-optimization over spectral width W .

Figure 5: Sensitivity results already present at the pinned commit.

231 6. Discussion

232 6.1. Engineering interpretation

233 The 8.26% and 3.37% reductions are case-specific, model-based changes
234 under common quantities and unit rates. Their value lies in the traceable
235 mechanism. M-C is 0.44% shorter, but the 261.4-million-RMB structure re-
236 duction is much larger than the cost implied by 98.3 m of length alone. The
237 optimizer trades more earthwork and a slightly higher slope-hazard diagnos-
238 tic for lower structure exposure and operating energy.

239 The sequential comparison reinforces the priority of feasibility. Its lower
240 raw objectives cannot be interpreted as a better design because $R_{min} =$
241 397.06 m violates the 400-m threshold. Collaborative search retains the abil-
242 ity to change plan and profile together near a constraint boundary; the pinned
243 record shows a feasible joint alternative where the particular sequential run
244 is infeasible.

245 6.2. Role of spatial data and IJS

246 Candidate-specific DEM sampling changes the objective landscape be-
247 cause lateral movement changes ground elevation rather than only distance.
248 OSM intersections convert location into bridge demand, while the connected
249 mountain mask makes ecological-tunnel length endogenous. These mecha-
250 nisms improve auditability but remain planning abstractions: OSM lacks
251 reliable deck elevations and parcel attributes, and the reconstructed DEM is
252 not a survey-grade pre-construction surface.

253 Tent, Lévy and DE play plausible coverage, exploration and refinement
254 roles. Among the single-operator variants, JS+DE has the largest observed
255 same-generation gap from JS; this observation does not isolate an evaluation-
256 budget-normalized DE effect. The algorithm evidence therefore supports the
257 implemented workflow and generates an equal-NFE research hypothesis; it
258 does not isolate an intrinsic 25.45% advantage.

259 6.3. Limitations and validity boundary

- 260 (1) The vertical start and end elevations are free. In the selected M-C, the
261 terminal elevation is visibly below the existing endpoint. This can affect
262 both earthwork and one-direction energy; the 8.26%/3.37% result must
263 not be presented as a fixed-tie-in design.

- 264 (2) Energy is evaluated for one representative direction. A bidirectional,
 265 direction-weighted model with calibrated regenerative braking is required
 266 to remove net-elevation sensitivity.
- 267 (3) Adjacent grade change is only a proxy for vertical-curve design. Explicit
 268 crest/sag radii, curve lengths, sight distance, clothoid/arc segmentation
 269 and full plan-profile coordination checks are absent.
- 270 (4) The quasi-natural DEM is reconstructed from a built-road surface at
 271 about 8.8-m resolution. OSM coverage and crossing classifications are
 272 heterogeneous, and obstacle vertical clearances are not verified.
- 273 (5) Building density does not enter this pinned model. Urban avoidance
 274 shown in conceptual graphics must not be interpreted as a computed
 275 constraint or benefit.
- 276 (6) Cost rates are planning estimates. Geology, hydrology, parcel acqui-
 277 sition, resettlement, construction staging, embodied carbon and uncer-
 278 tainty require project-specific data.
- 279 (7) The engineering solution is one optimization run. Ablation and bench-
 280 marks use same generations but unequal NFE; their rank-sum tests are
 281 unadjusted. Algorithm superiority therefore remains unconfirmed under
 282 equal budget.

283 Accordingly, the framework provides auditable corridor-planning evidence
 284 and a reproducible computational case. It does not replace detailed geometric
 285 design or support a universal saving rate.

286 7. Conclusions

287 This study links candidate three-dimensional highway geometry to quasi-
 288 natural terrain, OSM-triggered structures, life-cycle cost and mixed fuel-
 289 electric traffic energy. A 275-variable collaborative representation is searched
 290 by IJS, and feasibility, budget and non-dominance precede entropy-based
 291 compromise selection.

292 At the pinned `c1327df` version, the feasible Guangzhou North Ring joint
 293 solution reduces cost by 8.26% and monetized energy by 3.37%, shortens the
 294 line by 0.44%, and satisfies the radius requirement with $R_{min} = 401.03$ m
 295 and zero penalty. The principal economic mechanism is a 261.4-million-RMB

296 structure reduction that exceeds the 45.5-million-RMB earthwork increase. A
297 lower-objective two-stage result is rejected because its radius is 397.06 m. IJS
298 has the lowest observed same-generation objective in the stored ablation and
299 benchmark, but unequal NFE prevents an equal-budget superiority claim.

300 The most important next methodological correction is to anchor both
301 vertical endpoints and evaluate bidirectional traffic before treating the align-
302 ment as design-grade. Equal-NFE multi-seed algorithm comparison, explicit
303 vertical curves and survey/parcel data are also needed. These are limitations
304 and future work, not supplementary experiments used to alter the results
305 reported here.

306 **Data and code availability**

307 All manuscript numbers and manuscript-only figures are tied to Git com-
308 mit `c1327df`; its full hash is stored in the source manifest. The draft package
309 contains a source-value snapshot, SHA-256 manifest and a figure script that
310 reads the immutable Git objects without executing an optimizer. GNSS and
311 project-derived data may be subject to owner restrictions. A public reposi-
312 tory DOI should be added after anonymization review.

313 **CRedit authorship contribution statement**

314 **[Author placeholders]:** Conceptualization, Methodology, Software,
315 Validation, Investigation, Data curation, Visualization, Writing—original
316 draft, Writing—review & editing. Replace with verified author roles before
317 submission.

318 **Declaration of competing interest**

319 The authors declare that they have no known competing financial in-
320 terests or personal relationships that could have appeared to influence the
321 work reported in this paper. Confirm this statement with all authors before
322 submission.

323 **Acknowledgements**

324 Funding agency, grant number and non-author contributions are
325 anonymized and must be completed before submission.

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